

EE336 Fundamental of Photovoltaics, Lab I

Simulation Exercise 1 (20 – 30 min)

Go to <http://www.pveducation.org/pvcdrom/properties-of-sunlight/sun-position-calculator> and <https://www2.pvlighthouse.com.au/calculators/solar%20path%20calculator/solar%20path%20calculator.aspx> and familiarize with the simulation.

Key in different parameters (e.g., location, date, time and solar panel position) and observe the zenith angle, azimuth angle, air mass, incident angle to the solar panel and the solar path change with location, date, time and solar panel position.

Longitude, latitude and time zone of locations all over the world are available at www.timeanddate.com.

Simulation Exercise 2 (20 – 30 min)

Go to <http://www.pveducation.org/pvcdrom/properties-of-sunlight/calculation-of-solar-insolation>.

Simulate how the solar insolation varies with latitude position and day number of the year.

Simulate how the incident power, the power on the horizontal and module power varies with latitude position, tilt angle and day number of the year.

Simulate how the sunrise time, sunset time and number of sun hours varies with number of days.

Quiz 1.

A solar panel is installed parallel to the ground in Shenzhen as illustrated in Figure 1a. Assuming that there is no effect of the atmosphere and weather on the sun light intensity,

- Plot the sun position throughout the day on the polar plot for today.
- On what day and month of the year and what time does the solar panel receive the highest power from the sun?
- If the solar panel is tilted 30° from horizontal (Figure 1b) at that time, what is the relative change in the power received by the solar panel?
- Plot the sun's position throughout the day on the polar plot on that day.
- How should position of the solar panel be adjusted (in terms of tilt angle and azimuth angle) at 9am, 12am, 3pm and 5pm of that day to receive the maximum power from the sun?

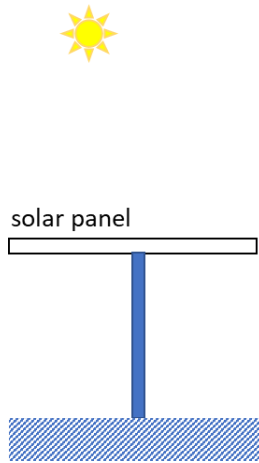


Figure 1a

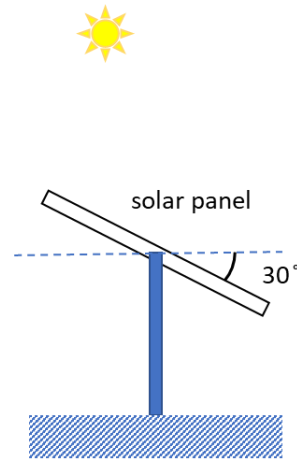


Figure 1b

Quiz 2.

A solar panel is installed with tilt angle 30° (as in Figure 1b) at Sydney (Australia).

- Plot the sun position throughout the day on the polar plot for today.
- If the azimuth angle is set to 0° , plot the air mass and incident angle to the panel versus hour of the day.
- Compare with the sun position plot that you observed in Shenzhen and comment on the observation.
- What should be the azimuth angle range of the solar panel for today?
- What is the optimized tilt angle and azimuth angle of the solar panel to receive the highest power from sun at today noon?

Quiz 3.

Two solar panels (A and B) are installed in New York City. The solar panel A is installed at tilt angle 0° and the solar panel B is installed at tilt angle 65° . Ignoring the local weather effects,

- Compare the solar insolation at New York City on 1st Jan and 30 June and calculate the solar intensity ratio between the two dates.
- On which day(s) does the solar panel A receive the highest power from the sun?
- On which day(s) does the solar panel B receive the highest power from the sun?
- Plot the solar power received by each solar panel through out the year. In which period of the year, the solar panel A outperform the solar panel B? In which period of the year, the solar panel B outperform the solar panel A?
- Should the tilt angle of the solar panel B be adjusted more than 65° ?

Quiz 4.

Suppose that you install a solar panel at the north pole and south pole.

- (a) Plot the sun position throughout the day on the polar plot for today.
- (b) Where is the sun position at mid night and noon?
- (c) Will you install the solar panel in horizontal position or vertical position?
- (d) At which tilt angle does the solar panel receive the lowest solar power?
- (e) How should the tilt angle of the solar panel be adjusted to receive the highest total solar power over the whole year?